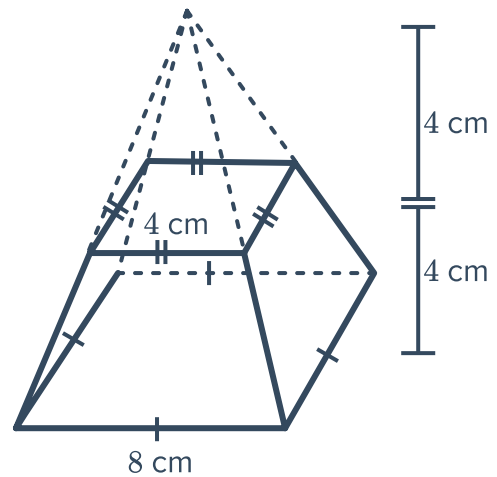


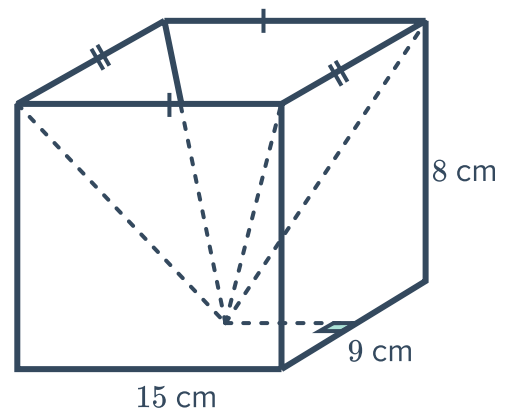
Volume of truncated pyramids and cones

- 1 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown.

Find the exact volume of the solid.

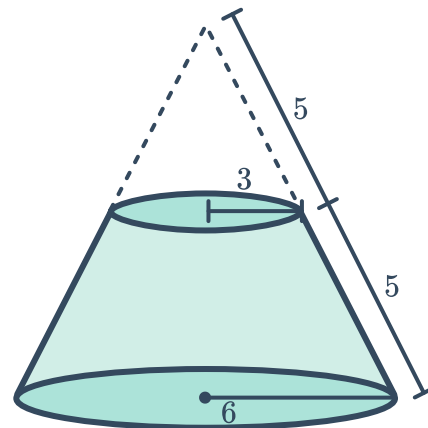


- 2 A pyramid has been removed from a rectangular prism, as shown. Find the volume of the remaining solid.



- 3 The top of a solid cone was sawed off to form the solid attached, such that radius of the top face is 3 cm and the radius of the bottom face is 6 cm.

Find the volume of the solid formed. Round your answer to two decimal places.

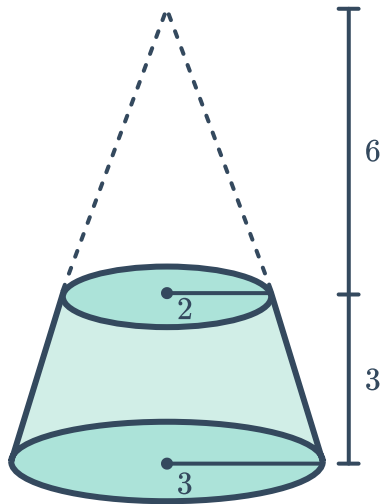


4 Each of the following solid is a truncated cone formed by cutting off a cone shaped section from the top of a larger, original cone.

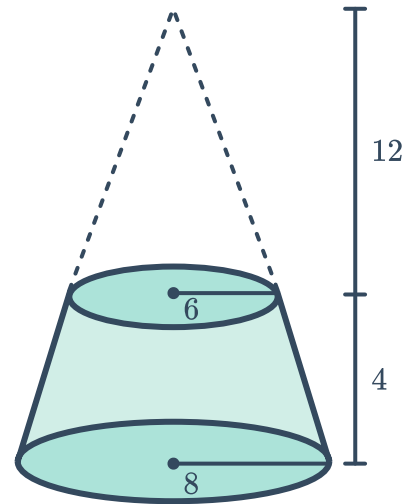


- i Find the exact volume of the cone section that was cut off.
- ii Find the exact volume of the original cone.
- iii Hence, find the volume of the truncated cone, correct to two decimal places.

a



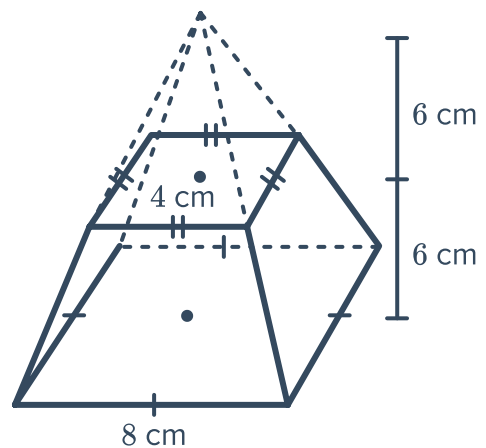
b



Surface area of truncated pyramids and cones

5 A small square pyramid of height 6 cm was removed from the top of a large square pyramid of height 12 cm to form the solid shown.

- a Find the length of the slant height of the sides of the new solid, correct to two decimal places.
- b Hence, find the surface area of the solid formed, correct to one decimal place.

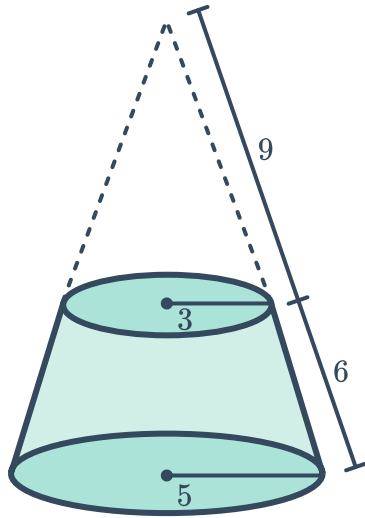


6 Each of the following solid is a truncated cone formed by cutting off a cone shaped section from the top of a larger, original cone.

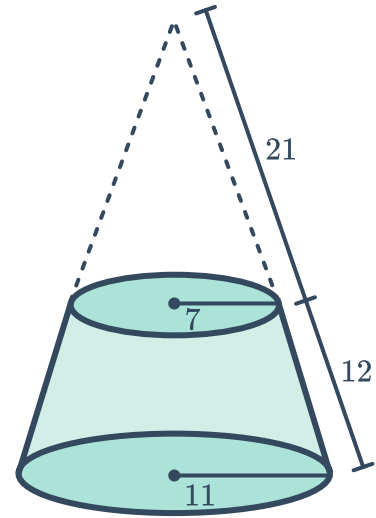


- i Find the exact curved surface area of the original cone.
- ii Find the exact curved surface area of the cone section that was cut off.
- iii Hence, find the surface area of the truncated cone, correct to two decimal places.

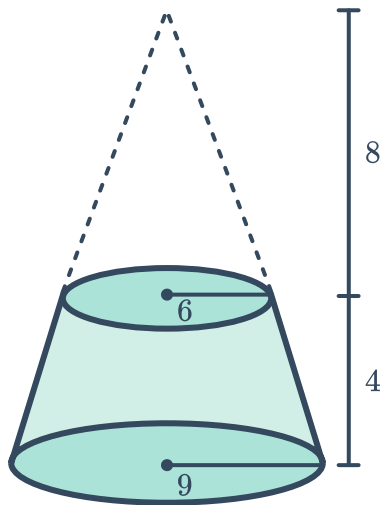
a



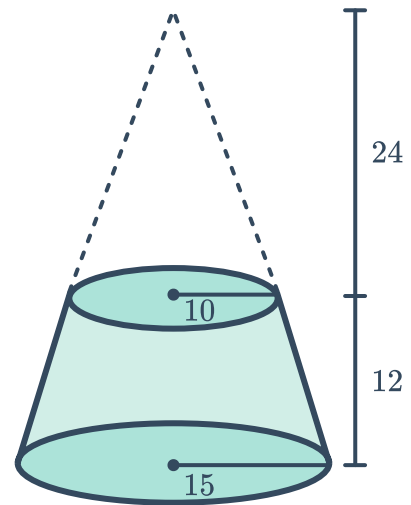
b



c



d

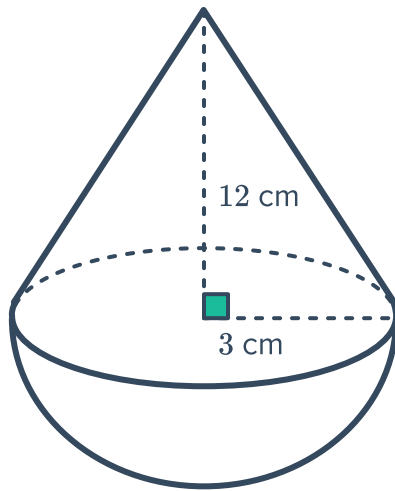




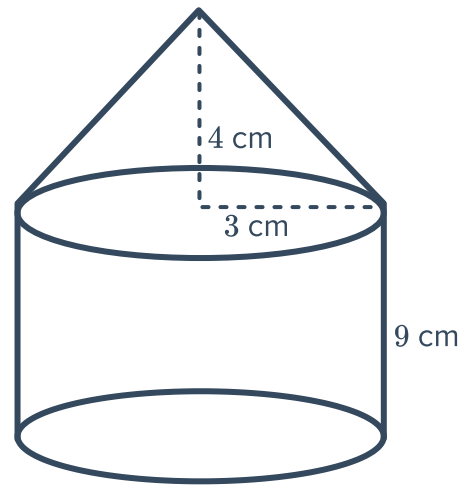
Other compound solids

7 Find the volume of the solids correct to two decimal places:

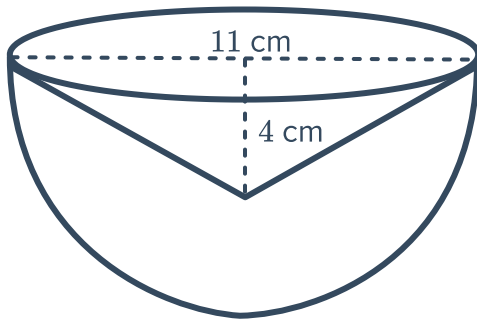
a



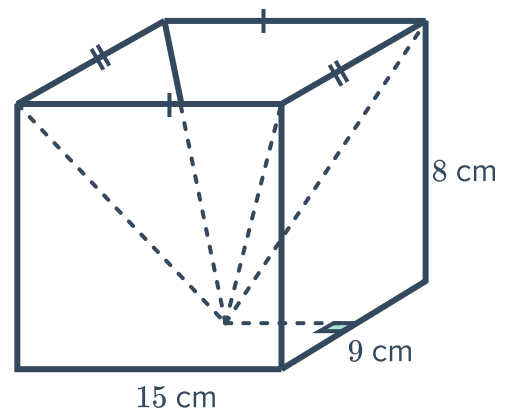
b



c

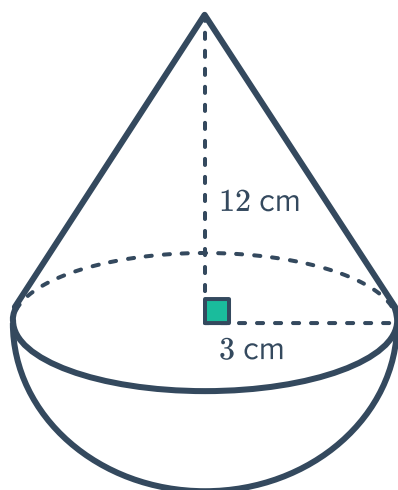


d

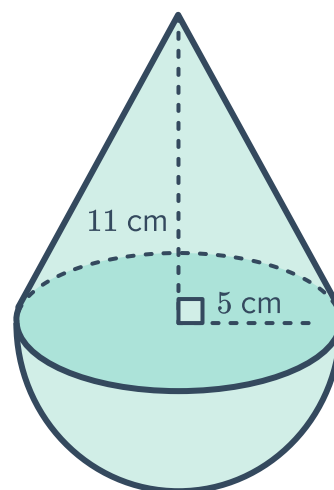


8 Find the surface area of the following composite figures, correct to two decimal places:

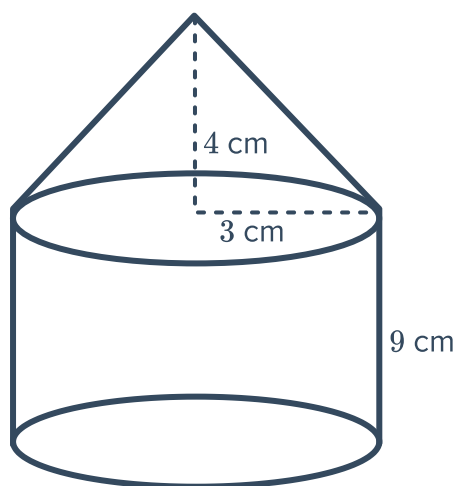
a



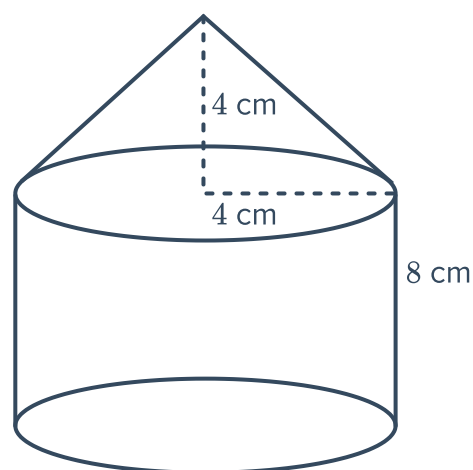
b



c



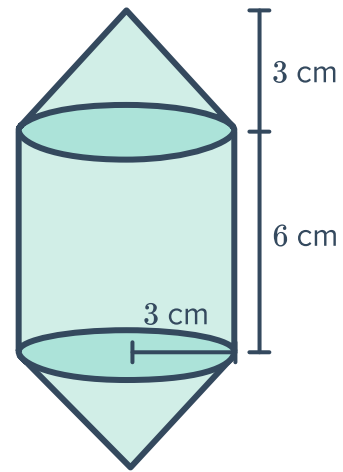
d



- 9 The figure shows a cylinder of radius 3 cm and its height is double the radius. On the top and bottom of the cylinder are cones with radii and height both also equal to 3 cm .

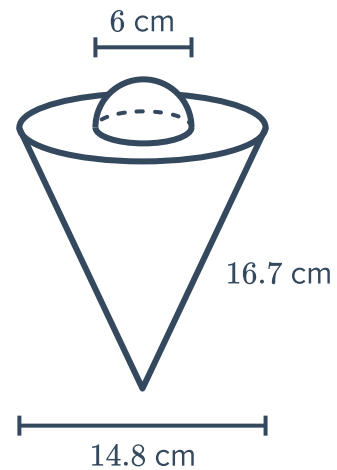


- a Describe the steps involved in calculating the surface area of this compound solid?
- b Find the surface area of the solid. Round your answer to two decimal places.



- 10 Consider the following solid:

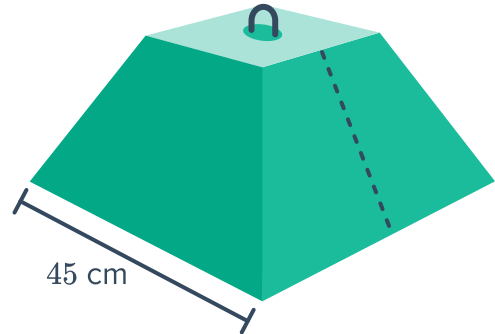
- a Find the perpendicular height of the cone, correct to one decimal place.
- b Find the volume of the solid, correct to two decimal places.



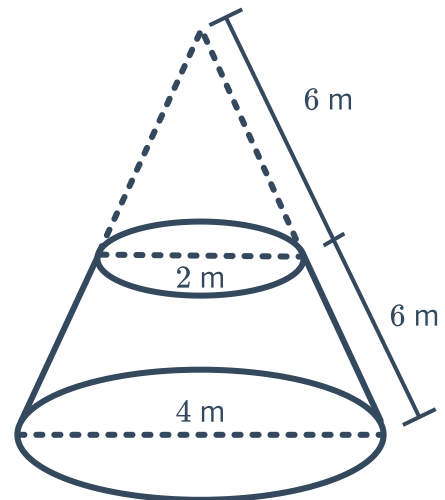


Applications

- 11 A weight is constructed by removing the top 38 cm from a 57 cm tall pyramid with a square base of side length 45 cm.
- Find the volume of the original pyramid.
 - Calculate the side length of the square on top of the weight.
 - Calculate the volume of the removed part of the pyramid.
 - Calculate the volume of the weight.



- 12 The following podium was formed by sawing off the top of a cone. Find its volume, correct to two decimal places.



13 Before 1980, Mount St. Helens was a volcano approximately in the shape of the top cone below:



a What was the volume of the mountain, in cubic kilometres? Round your answer to two decimal places.

b The tip of the mountain was in the shape of the bottom cone shown.

Find the volume of the tip in cubic kilometres. Round your answer to two decimal places.

c In 1980, Mount St. Helens erupted and the tip was destroyed.

Find the volume of the remaining mountain, in cubic kilometres. Round your answer to two decimal places.

